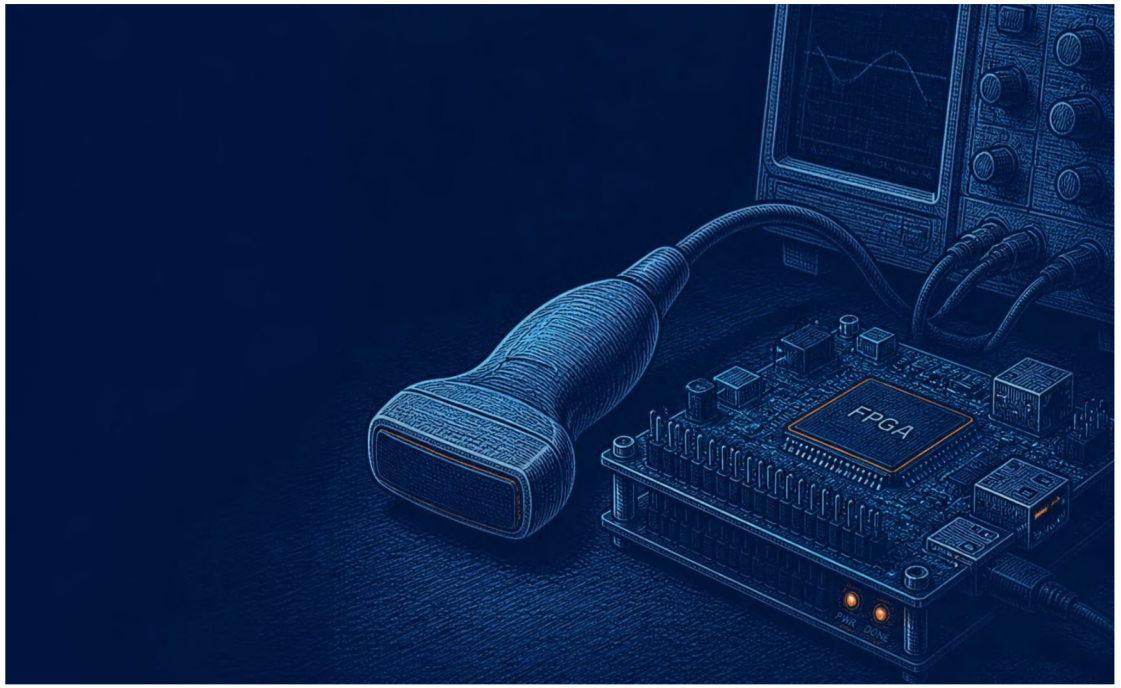
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सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY
सत्यमेव जयते

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NIELIT

FPGA Prototyping for Medical Ultrasound Imaging

From acoustic echoes to focused samples and hardware captures.

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Senior Technical Officer
National Institute of Electronics and Information Technology, Calicut



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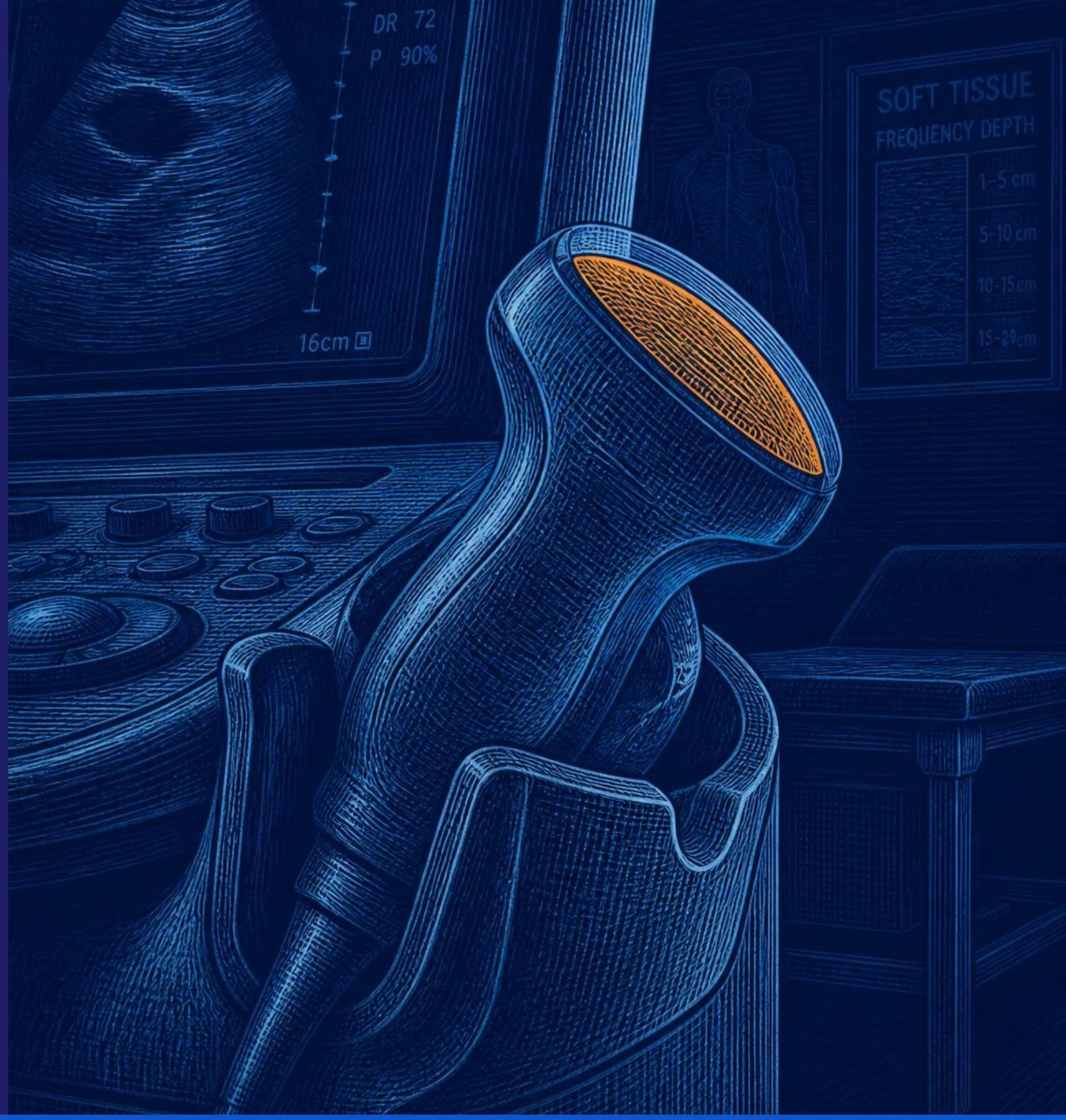
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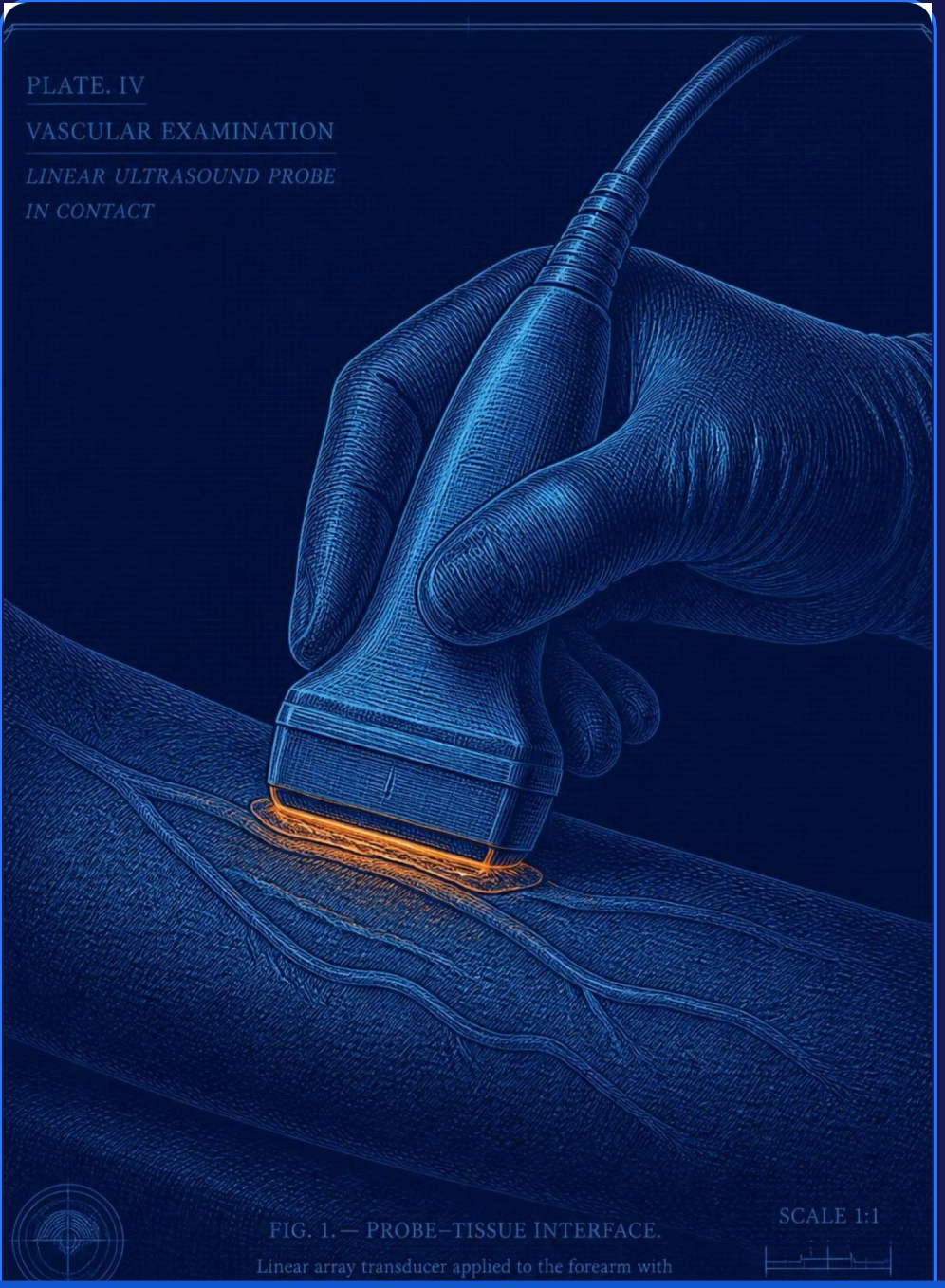
Debugging with ILA and VIO

Section 1

What is ultrasound imaging?

Pressure waves become spatial information.





ACOUSTICS → INFORMATION

Ultrasound imaging sees with sound

Ultrasound is sound above human hearing, over 20 kHz.

A probe sends pulses and listens for echoes. Their timing gives position; their strength contributes brightness.

Sound waves: frequency, wavelength, speed

$$\lambda = \frac{c}{f}$$

Frequency

Oscillations per second, in hertz.

Wavelength

Distance occupied by one cycle.

Speed

About 1540 m/s assumed in soft tissue.

Higher frequency gives finer detail, but less penetration.

Pulse-echo: distance from time



$$d = \frac{ct}{2} = \frac{1540 \times 130 \times 10^{-6}}{2} \approx 0.100\ \text{m}$$

About 10 cm deep. The pulse repetition interval must allow the deepest echo to return.

Medical ultrasound is real-time, safe imaging inside the body

Non-ionizing, portable, and generally lower cost than CT or MRI.

Used in cardiology, obstetrics, abdominal, vascular, and musculoskeletal imaging.

Frame rates range from tens to hundreds per second, depending on mode and depth.

Use appropriate output and exposure time. Non-ionizing does not mean unlimited acoustic power.



How tissue shapes the echo

Reflection

Changes in acoustic impedance send part of the wave back.

Scattering

Small structures redirect sound. Interference produces speckle.

Attenuation

Energy is lost with travel. Loss increases with depth and frequency.

Acoustic impedance describes how a medium relates pressure to particle motion.

Frequency versus depth trade-off

DEEPER TARGETS

2–5 MHz

Abdomen and heart

Lower attenuation supports greater depth.

SUPERFICIAL TARGETS

7–15 MHz+

Thyroid, vessels, skin

Shorter wavelengths support finer detail.

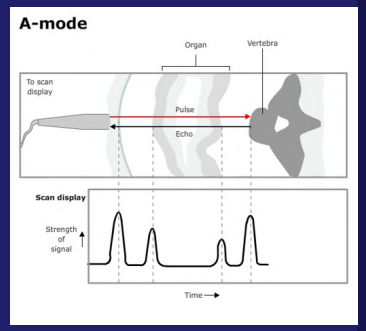
Probe choice sets front-end bandwidth, sampling, and delay precision requirements.

Imaging modes: A, B, M, and Doppler

A-MODE

Amplitude along a line

Echo strength versus depth.



B-MODE

Brightness in 2D

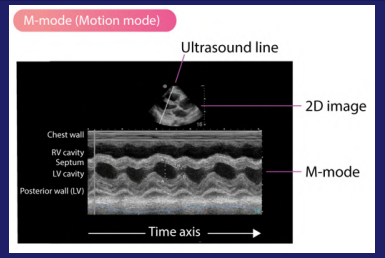
Grayscale anatomy assembled from scan lines.



M-MODE

Motion over time

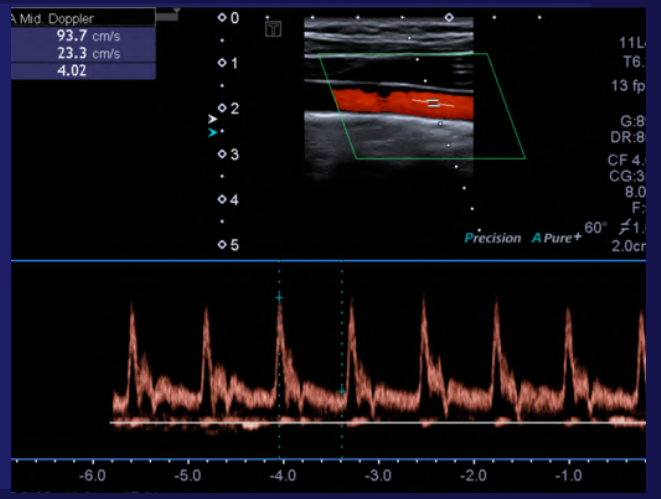
Repeated measurements along one line.



DOPPLER

Flow velocity

Color maps or spectral velocity displays.



PART 2

Radar, sonar, and medical ultrasound

Echo processing transfers. Propagation physics does not.

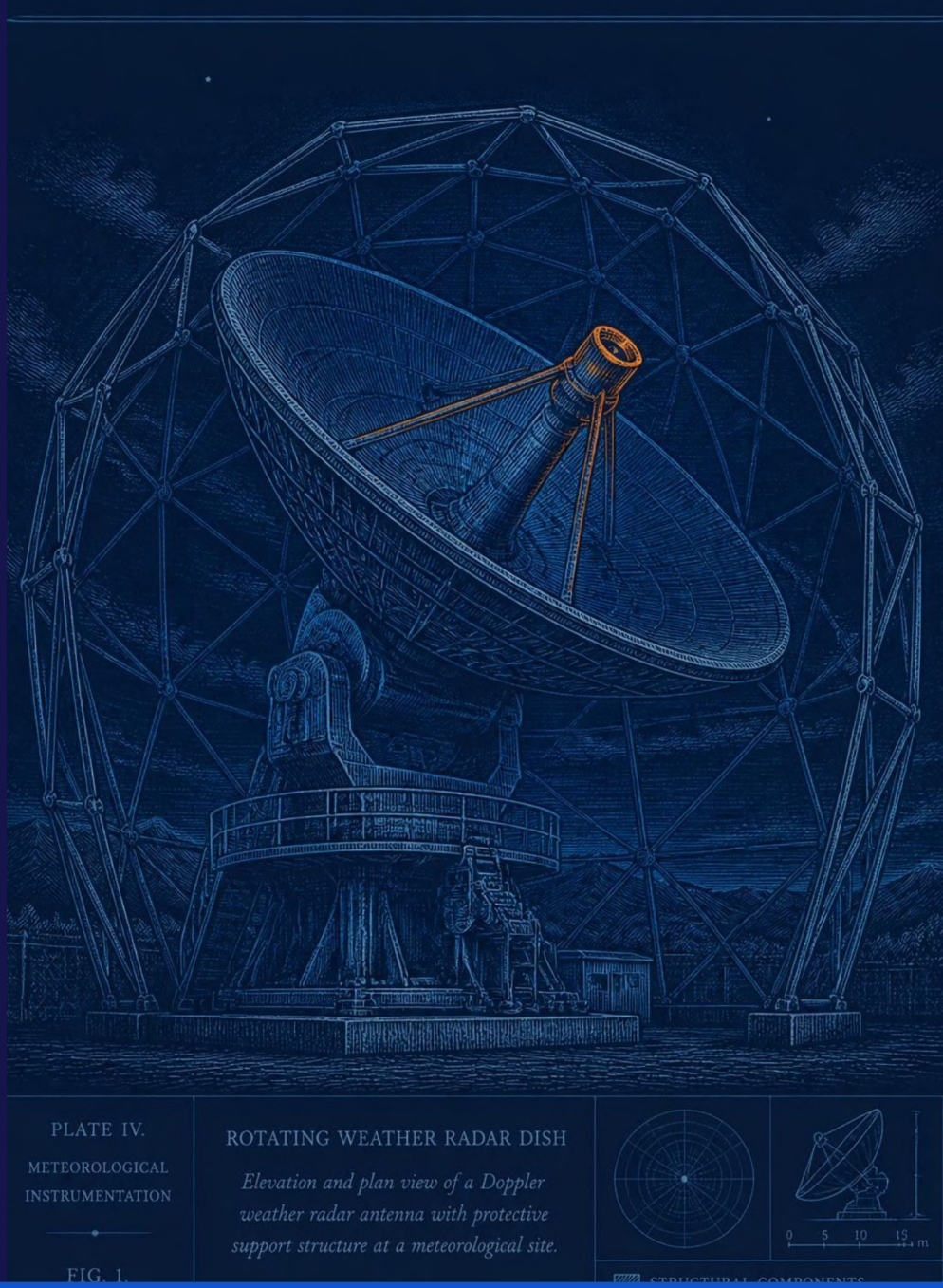


Radar: radio waves in air

Electromagnetic waves propagate at approximately the speed of light.

Aircraft, ships, and weather are detected across kilometres and beyond.

Phased arrays steer beams electronically.





Sonar: sound in water

Acoustic waves travel at about 1500 m/s in seawater.

Active sonar sends pulses; passive sonar only listens. Hydrophones are underwater acoustic receivers.

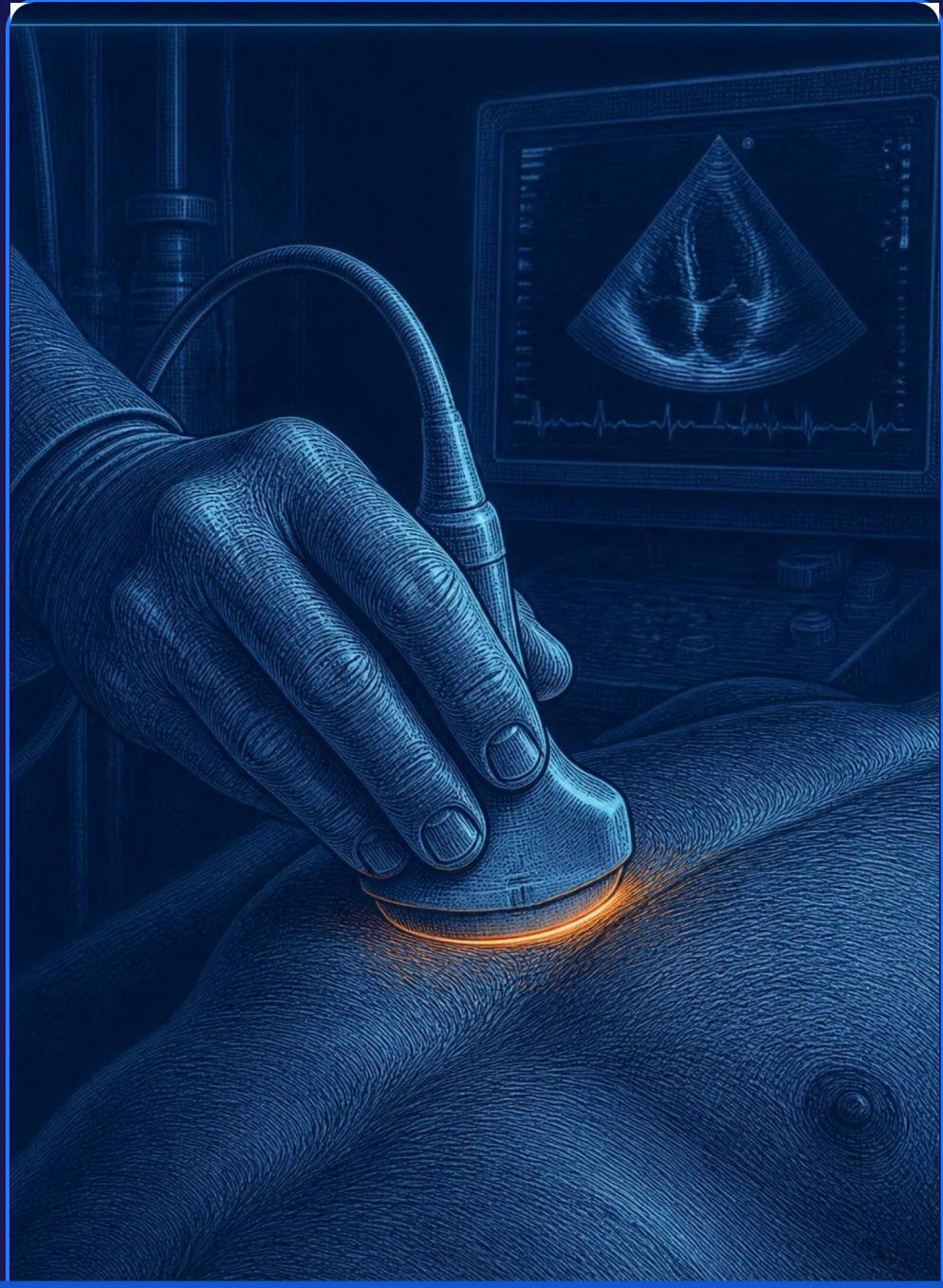
Operating ranges span metres to many kilometres.

Medical ultrasound: sound in tissue, centimetres away

Megahertz acoustics; about 1540 m/s assumed in soft tissue.

Centimetre depths, with resolution of a millimetre or less.

Real-time images from 64 to 256 or more receive channels.



The differences side by side

	Radar	Sonar	Medical ultrasound
Wave type	Electromagnetic	Acoustic	Acoustic
Medium	Air / space	Water	Soft tissue
Speed	~3x10^8 m/s	~1500 m/s	~1540 m/s
Range	Km and beyond	Metres to km	Centimetres
Goal	Detect, track	Detect, map	Image anatomy

Same idea, different physics and constraints

Transmit

All three send energy into a medium.

Wait

All three wait for echoes; propagation time encodes range.

Process

All three use arrays and beamforming to isolate direction.

Medical ultrasound adds strong attenuation, speckle, and a safety limit on power.

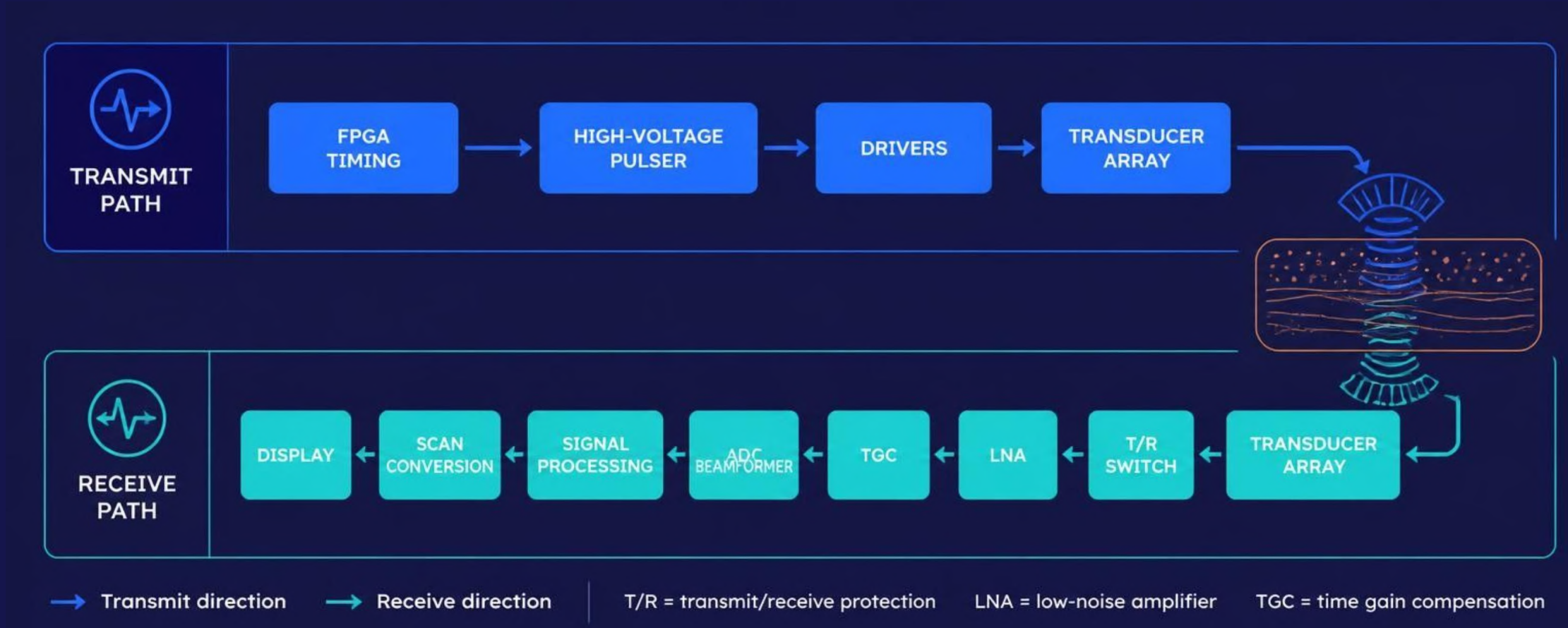
PART 3

Anatomy of an ultrasound machine

Follow the signal across analog, digital, and host boundaries.



The ultrasound system signal chain





The transducer: piezoelectric array

Piezoelectric elements convert voltage into pressure, and returning pressure into voltage.

Linear, convex, phased, and matrix probes serve different fields of view.

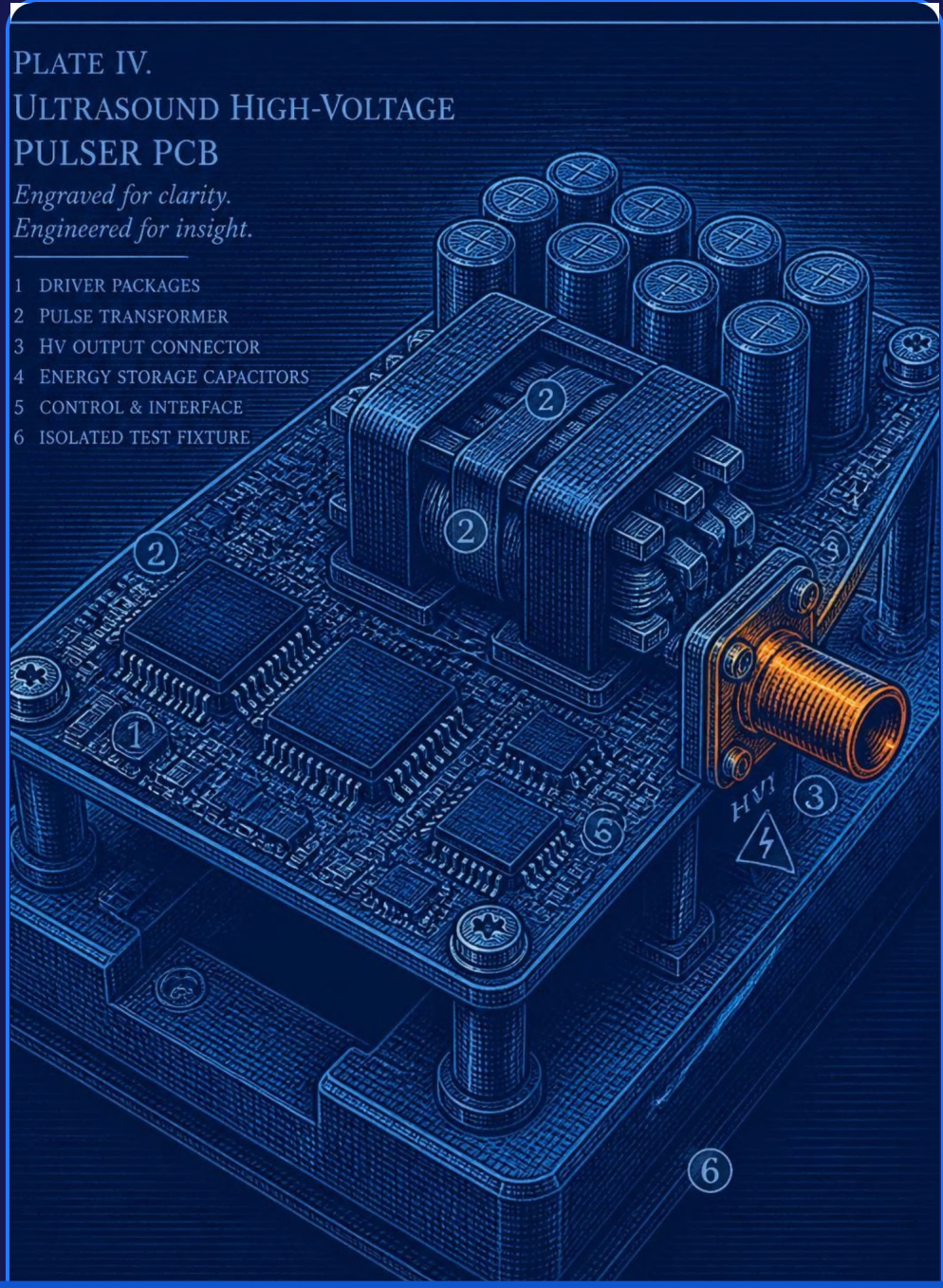
Typically 64 to 256 elements; 3D probes can contain thousands.

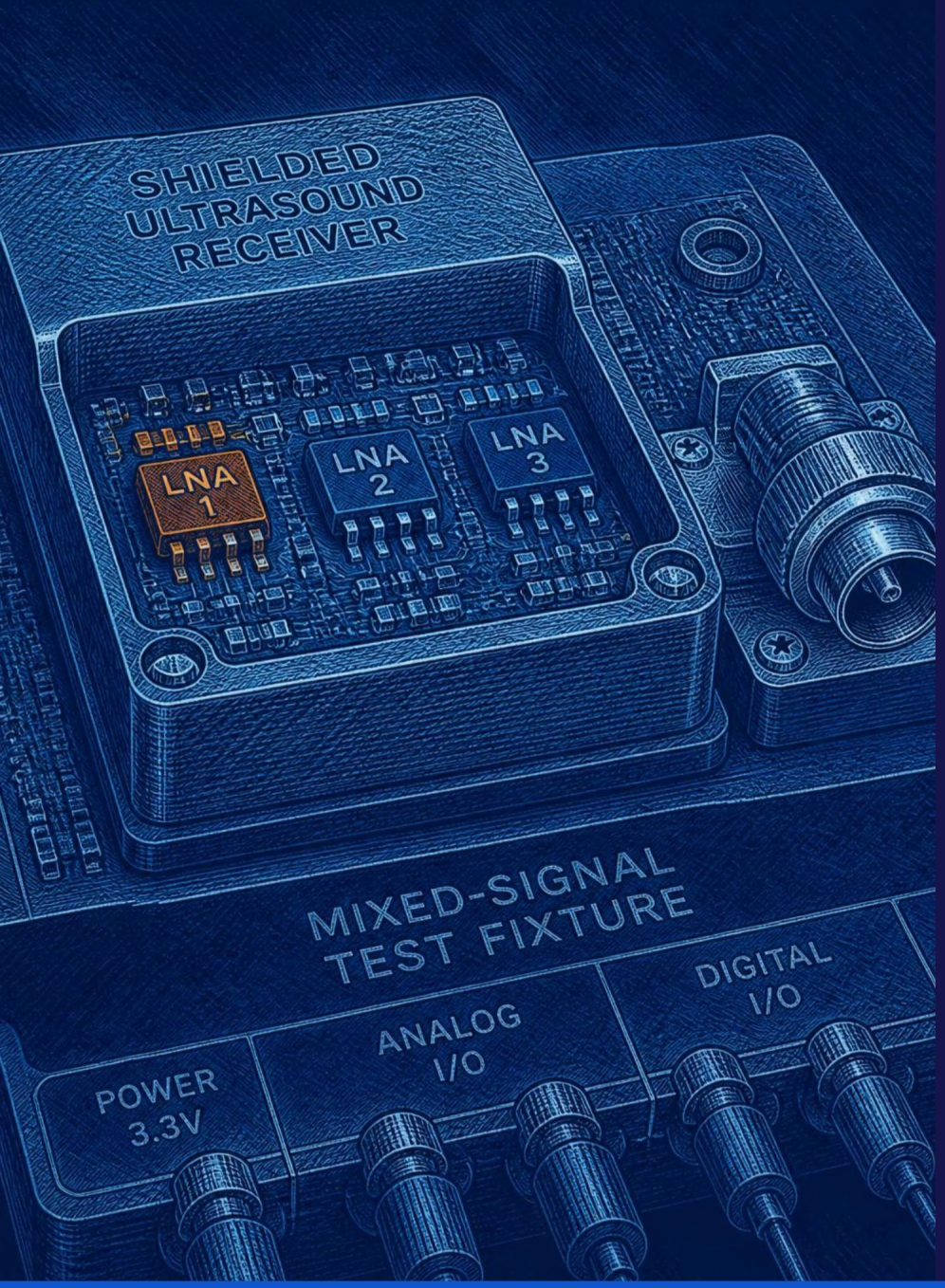
Transmit: pulsers and high voltage

Short excitation pulses reach tens of volts or more.

Per-channel firing delays set transmit focus and steering.

The FPGA generates precise timing patterns.





Receive front end: T/R switch and LNA

The transmit/receive switch protects sensitive receiver circuitry from the transmit pulse.

The low-noise amplifier handles microvolt echoes.

Noise introduced here sets the sensitivity floor for everything downstream.

Time gain compensation fights attenuation

01

Transmit ends

Start the depth-dependent gain schedule.

02

Early echoes

Shallow targets need less amplification.

03

Late echoes

Deeper targets need more gain.

An analog variable-gain amplifier, or FPGA-controlled gain schedule, compensates the expected loss.

Gain cannot recover signal already buried in noise.

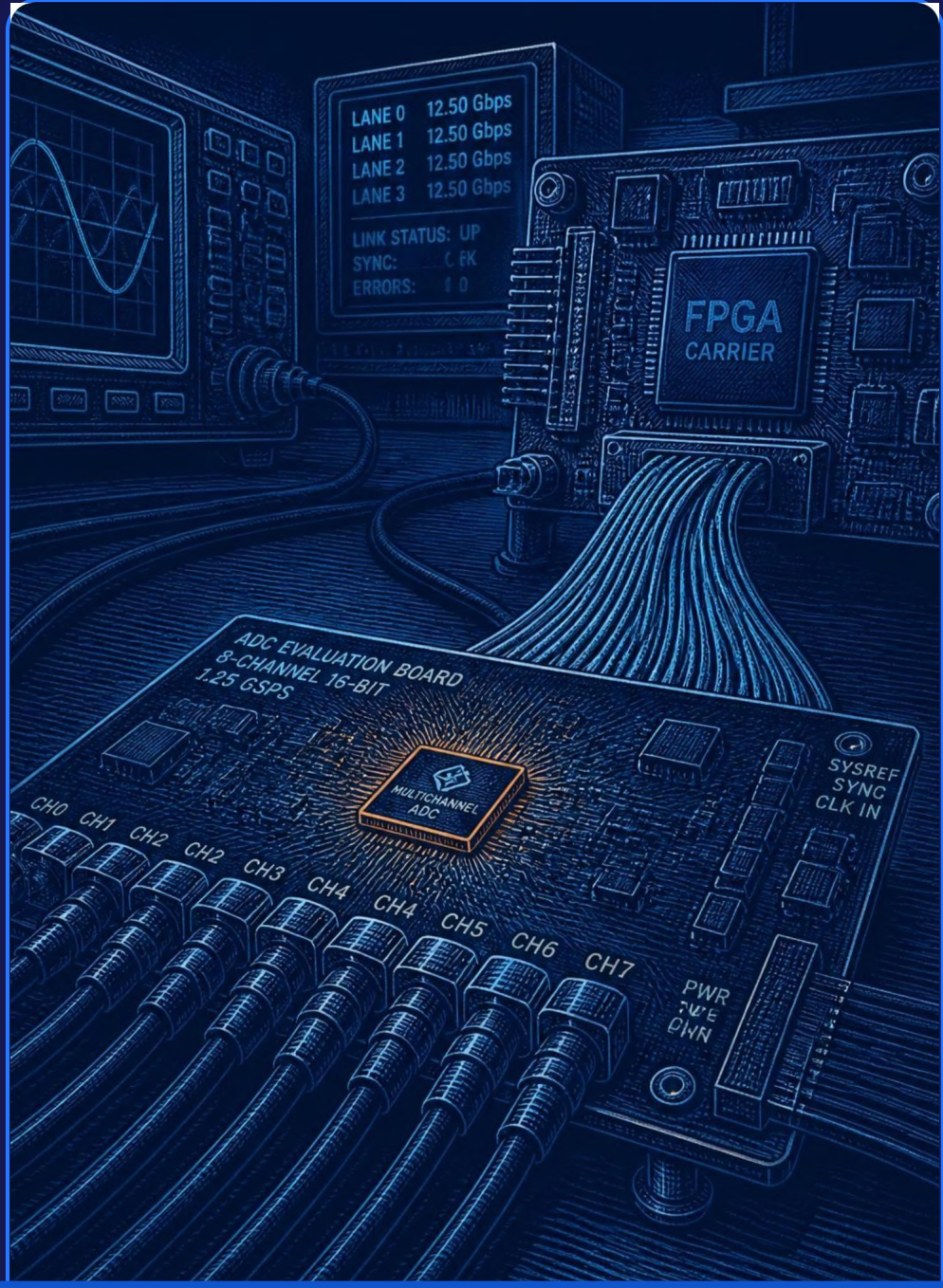
ADC: channels become samples

10–14 bits

Tens of MSPS per channel, with sampling several times the probe center frequency.

LVDS or JESD204 carries the digitized channels into the FPGA.

MSPS means million samples per second.



Beamformer: the heart of the machine

Focused line

Combines all channels into one focused line

Every scan

Repeats for every scan line, every frame

Compute-intensive

Most compute-intensive block in the chain

After the beamformer: from RF to brightness

01

Demodulate

Filter RF and form in-phase and quadrature signals, I/Q.

02

Detect

Compute the envelope, a measure of echo magnitude.

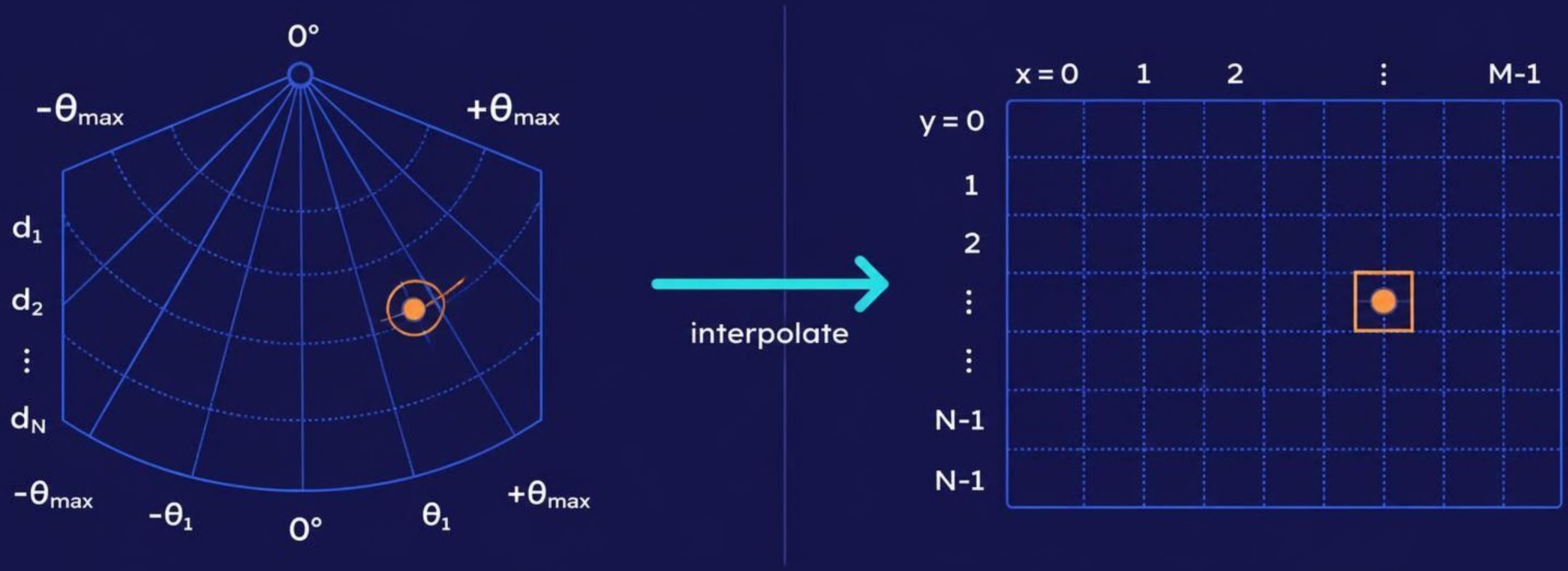
03

Compress

Log compression fits a large amplitude range into display brightness.

RF is the oscillating radio-frequency-band echo signal, before baseband conversion.

Scan conversion maps lines to pixels



Polar or line-coordinate samples are interpolated onto a display grid, often in a GPU or FPGA.

Doppler processing for flow

Repeat pulses

Observe phase changes at the same sample location.

Reject clutter

A wall filter suppresses slow tissue motion.

Estimate flow

Map the velocity component along the beam onto B-mode.

Doppler measures motion through echo frequency or phase changes, not brightness alone.

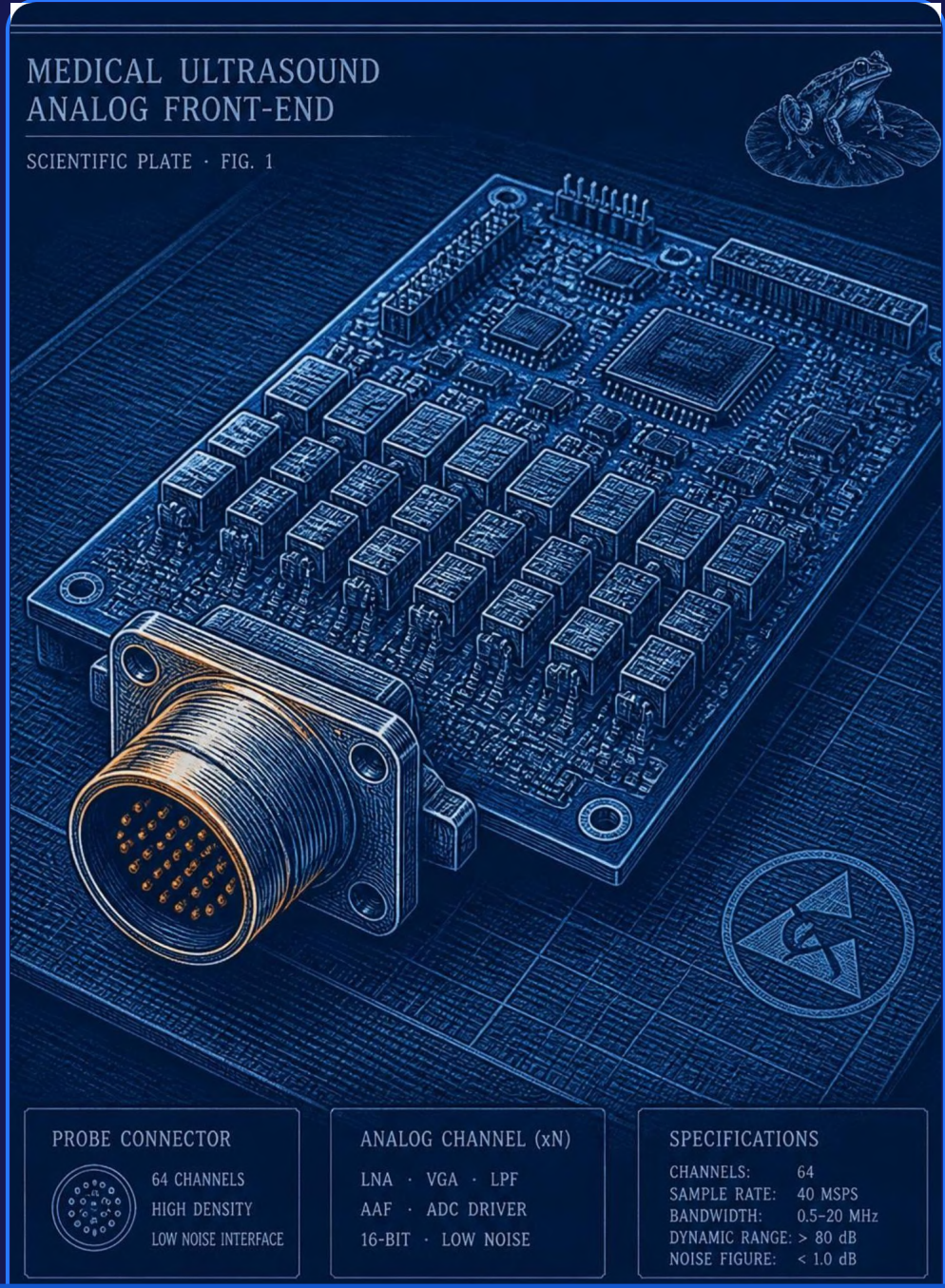
BOARD 1

Analog front end board

Pulsers, T/R switches, LNA, VGA, and ADCs sit close to the probe connector.

Mixed-signal layout and power noise directly affect echo quality.

VGA means variable-gain amplifier.



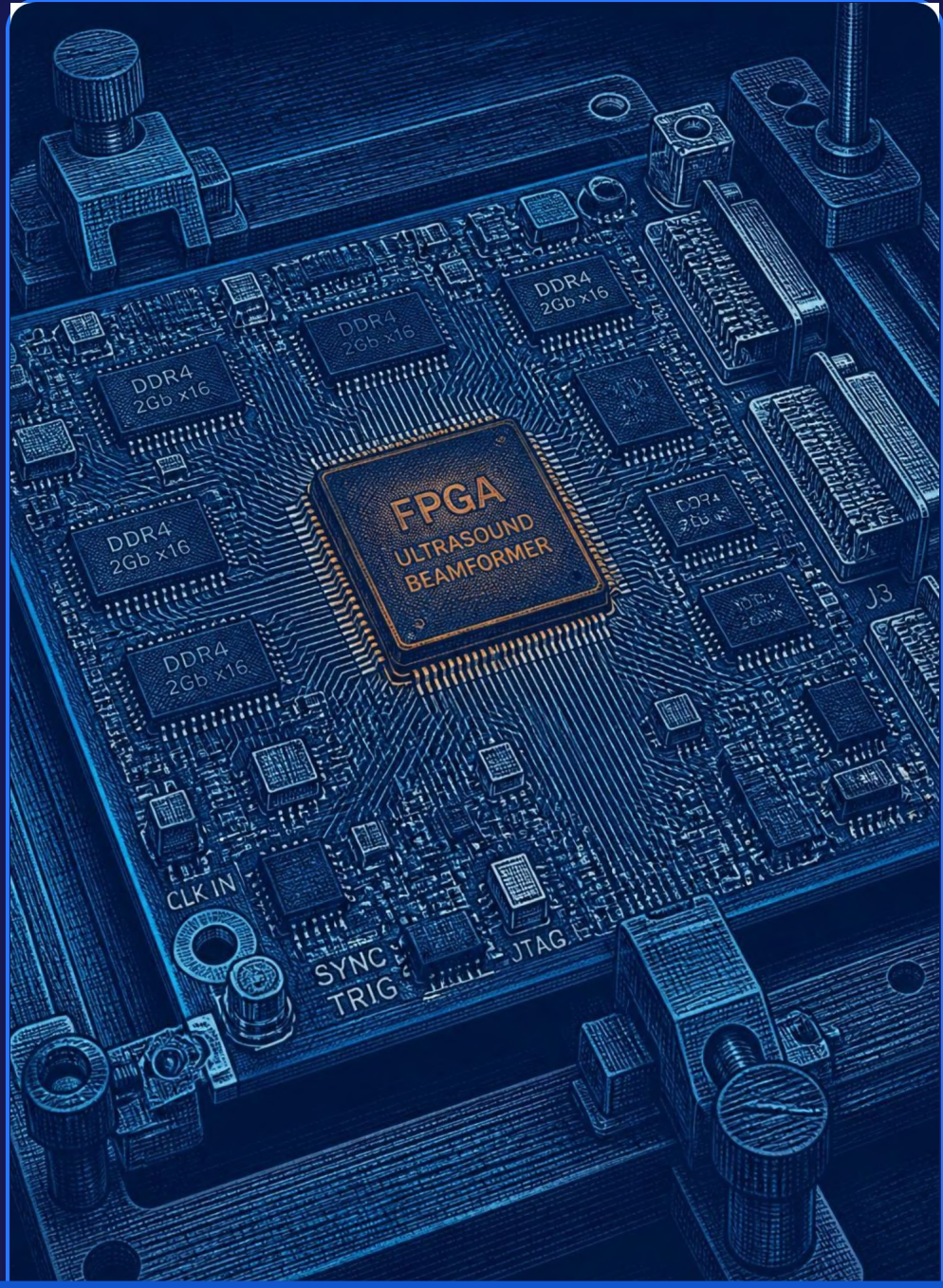
BOARD 2

Digital beamformer board with FPGA

Receives ADC streams, applies delays, and sums channels.

A mid- or high-end FPGA or SoC provides parallel compute and memory.

Beamformed lines leave for the host.



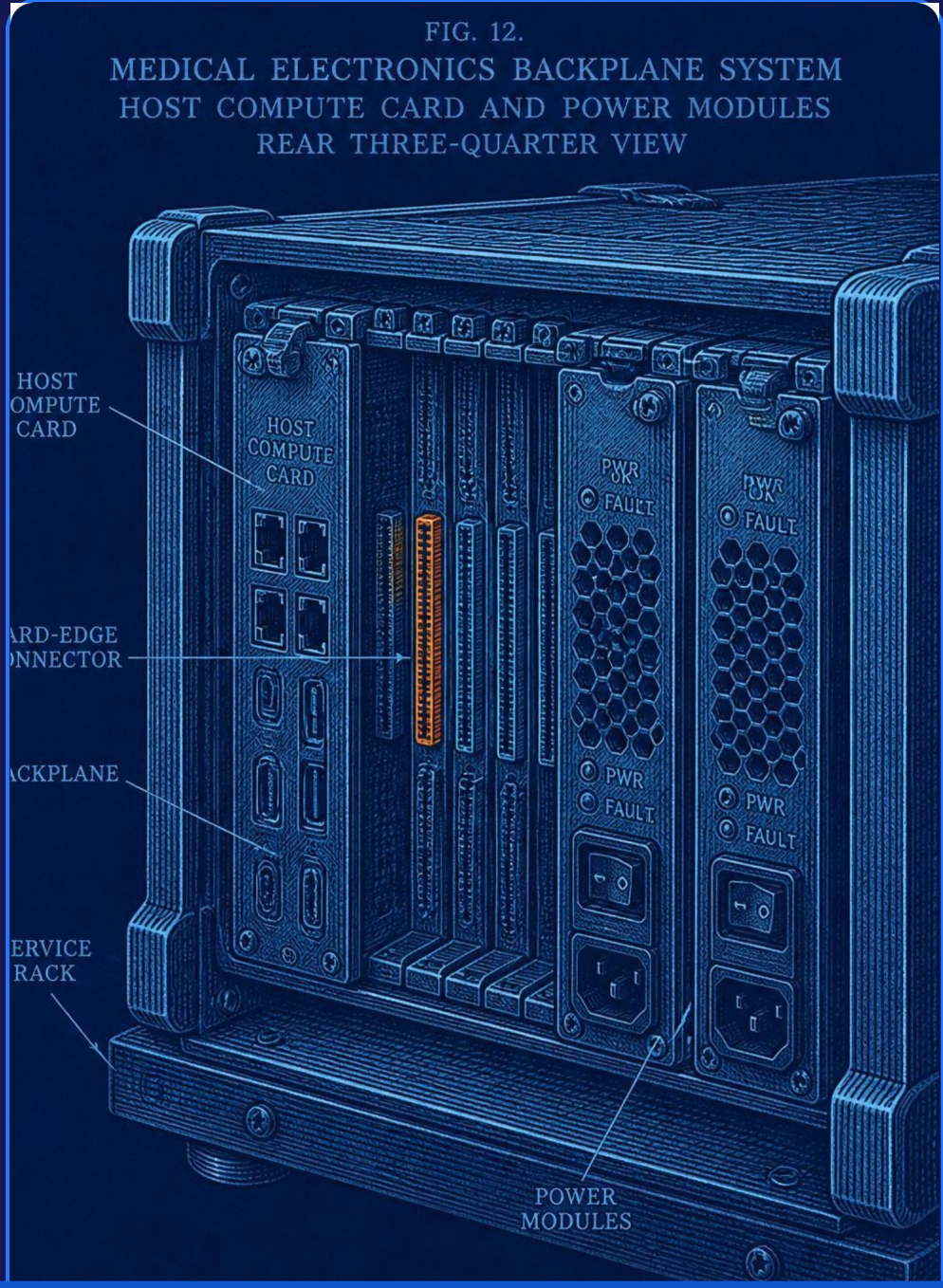
BOARD 3

Host, power, and backplane

CPU or GPU handles image processing and the user interface.

PCIe or Ethernet links the FPGA to the host.

High-voltage and low-noise supplies, a backplane, and probe multiplexing complete the platform.



Where the FPGA fits in the system

Timing

Transmit sequencing, per-channel delays, and synchronization.

Compute

Receive beamforming and high-rate filtering.

Transport

Bridge samples to the host while algorithms evolve.

Prototyping flexibility lives inside a deterministic streaming datapath.

PART 4

Beamforming basics

Geometry becomes timing. Timing becomes focus.



THE CENTRAL IDEA

Beamforming is why the image exists

One element hears broadly. An array can focus.

Combining elements gives direction and selectivity. The resulting beam shapes image contrast and resolution.

What is beamforming?

Align

Compensate each channel’s propagation time.

Add

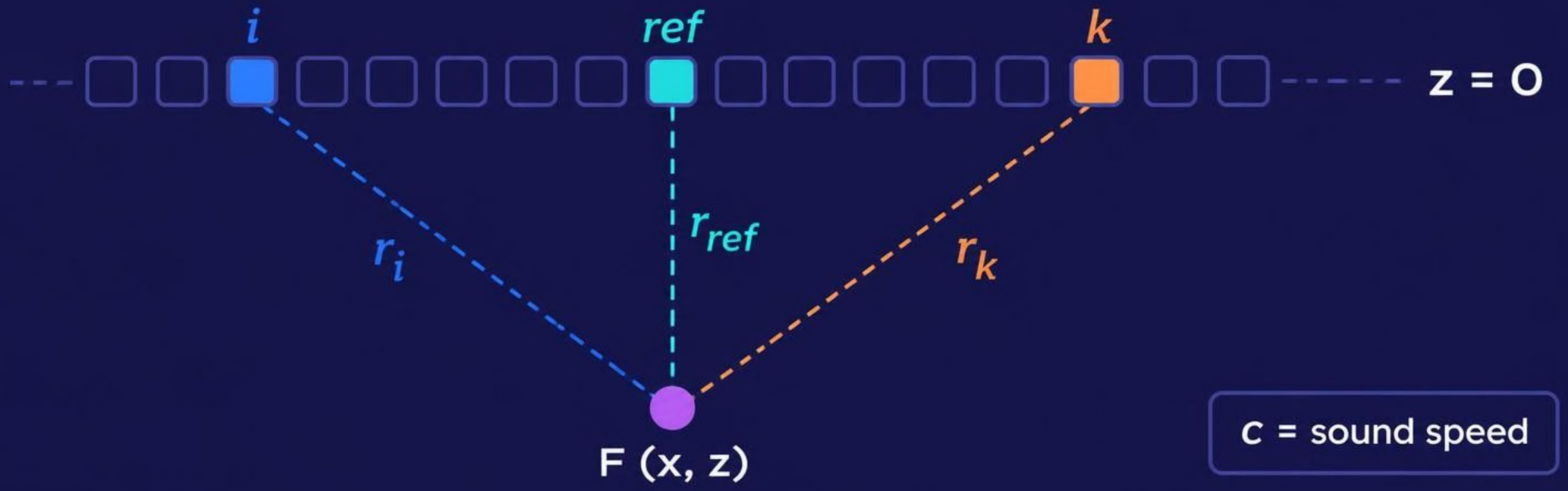
Echoes from the selected focus reinforce coherently.

Select

Off-focus echoes are less coherent and tend to cancel.

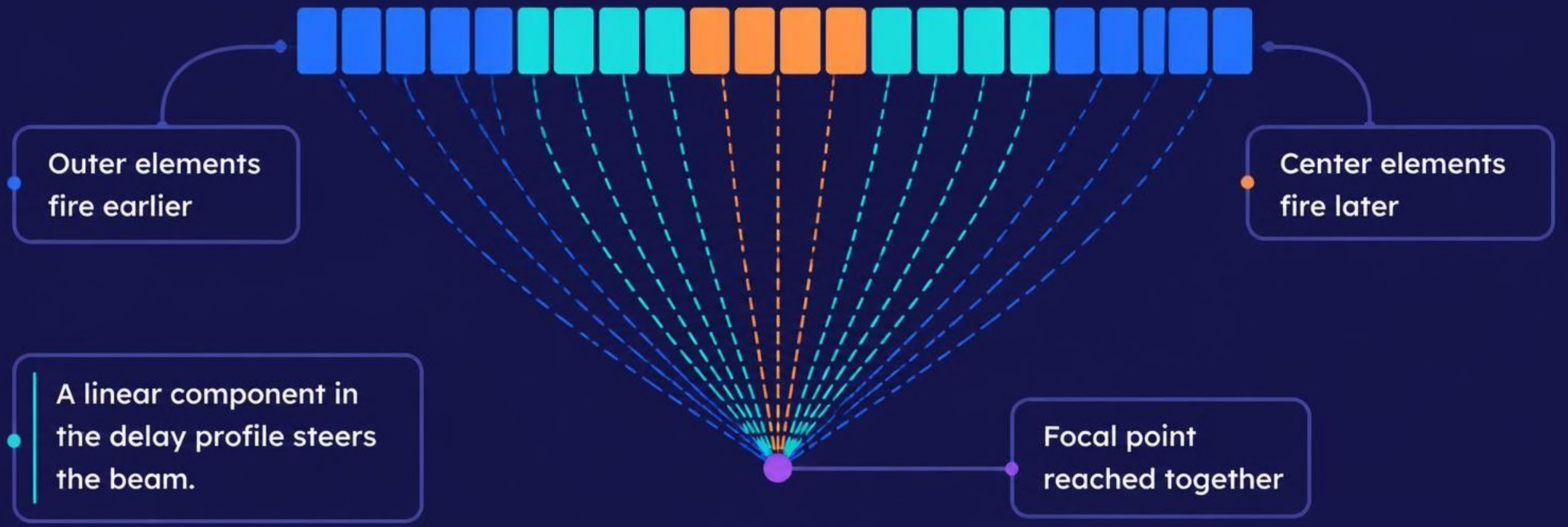
Coherent means aligned in phase, so contributions reinforce predictably.

Array geometry and path delays



$$\Delta t_i = \frac{r_i - r_{\text{ref}}}{c}$$

Transmit beamforming: focus on the way out



For an on-axis focus, longer outer paths require earlier firing.

Receive delay-and-sum

Delay

Align echoes associated with the focal point.

Weight

Apply apodization across the aperture.

Sum

Produce one focused RF sample.

$$y[n] = \sum_i w_i[n]\, x_i[n - d_i[n]]$$

Fractional sample delays require interpolation.

Computing the delay for each element

$$t_i = \left((x - x_i)^2 + z^2\right)^{1/2} / c$$

Focal point (x, z) ; element position x_i .

$$\Delta t_i = t_i - t_{\min}$$

Here, $t_{\min}$ is the minimum arrival time across elements. These relative arrival offsets describe geometry. A causal alignment buffer delays earlier arrivals by $\max_j(t_j) - t_i$.

Keep geometric arrival time distinct from the delay inserted by RTL.

Dynamic receive focusing

Shallow echo

Use the delay profile for a nearby focus.

Later echo

Advance the focus deeper into tissue.

Next sample

Update delays for the next depth.

Each scan line needs a depth-dependent delay profile, potentially updated sample by sample.

Apodization trades width for sidelobes

Uniform weights

All active elements contribute equally.

Narrower main beam, higher sidelobes.

Hann or Hamming

Edge elements contribute less.

Lower sidelobes, wider main beam.

Sidelobes are unwanted sensitivity away from the main beam. Weights become RTL multiplier coefficients.

Resolution: axial and lateral

ALONG THE BEAM

Axial

Shorter pulses separate nearby reflectors in depth.

$$\Delta z \approx \frac{\text{spatial pulse length}}{2}$$

ACROSS THE BEAM

Lateral

A larger aperture narrows the beam.

$$\Delta x \propto \frac{\lambda z}{D}$$

D is active aperture width. The proportionality depends on beam shape and the resolution criterion.

Element pitch, grating lobes, and delay precision

Pitch

Spacing near half a wavelength helps avoid grating lobes over useful steering angles.

Delay step

Coarse time steps introduce phase error and degrade sidelobes.

Fine delay

Oversampling or interpolation improves time alignment.

Grating lobes are additional strong beams caused by spatial sampling.

Worked example: 8-element delay-and-sum

Pitch p, on-axis focus z, sample rate f_s; symmetry reduces eight paths to four pairs.

Relative delay

$$d_i = f_s \frac{r_{\max} - r_i}{c}$$

Split each delay into integer and fraction. After alignment, identical unit-weight echoes sum to eight times one echo.

Four symmetric path pairs

Outer pair · position $\pm 3.5p$

Pair · position $\pm 2.5p$

Pair · position $\pm 1.5p$

Inner pair · position $\pm 0.5p$

Beyond line-by-line: modern beamforming schemes

Multi-line receive

Form several receive lines from one transmit event.

Plane waves

Compound broad transmissions for ultrafast imaging.

Synthetic aperture

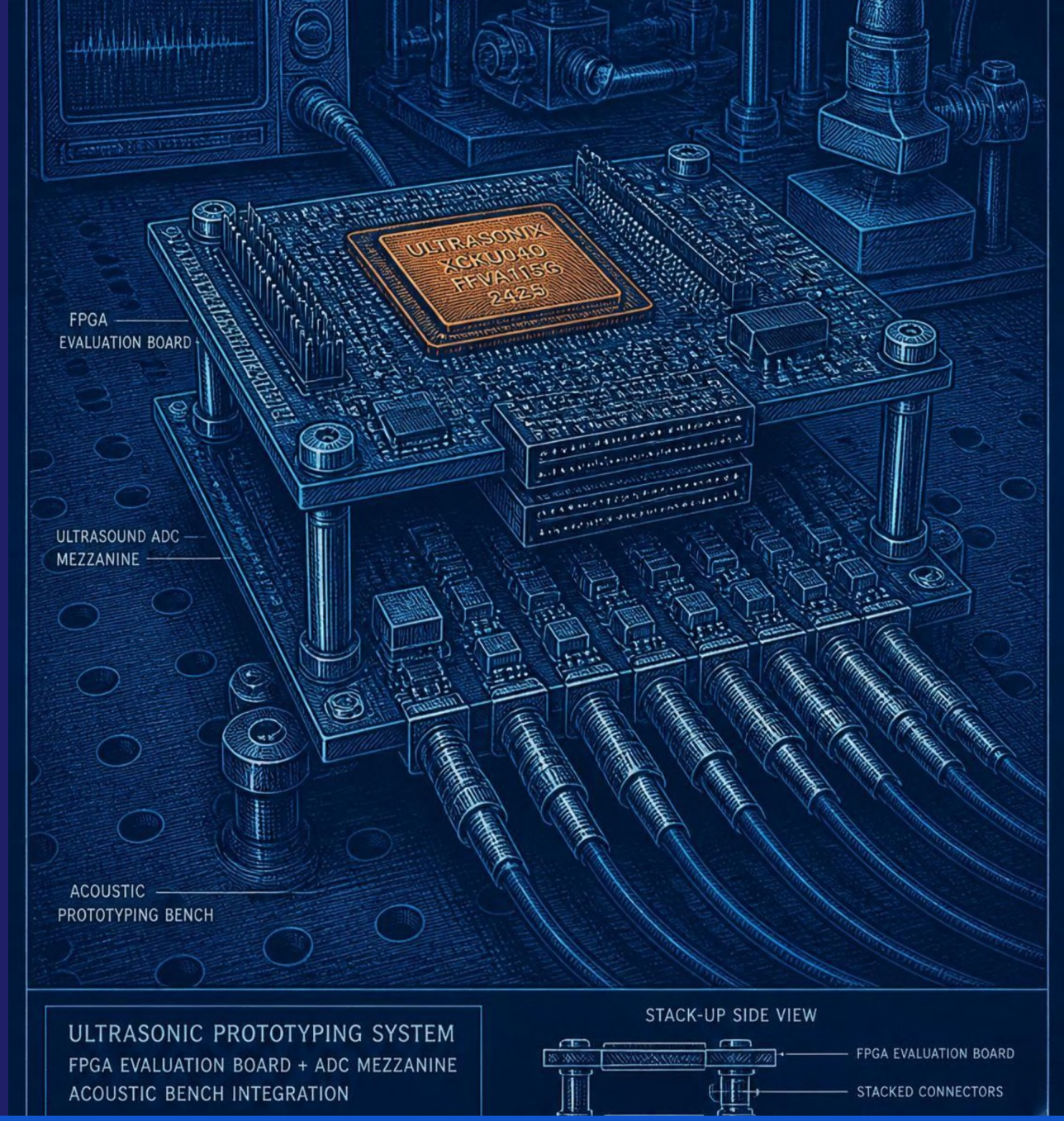
Combine acquisitions to synthesize a larger effective aperture.

The alignment math remains. Acquisition, storage, and bandwidth demands change.

PART 5

Receive beamforming on FPGA

Turn focal geometry into a streaming architecture.



Why FPGAs suit receive beamforming

Parallel lanes

Process channels concurrently rather than serially.

Predictable timing

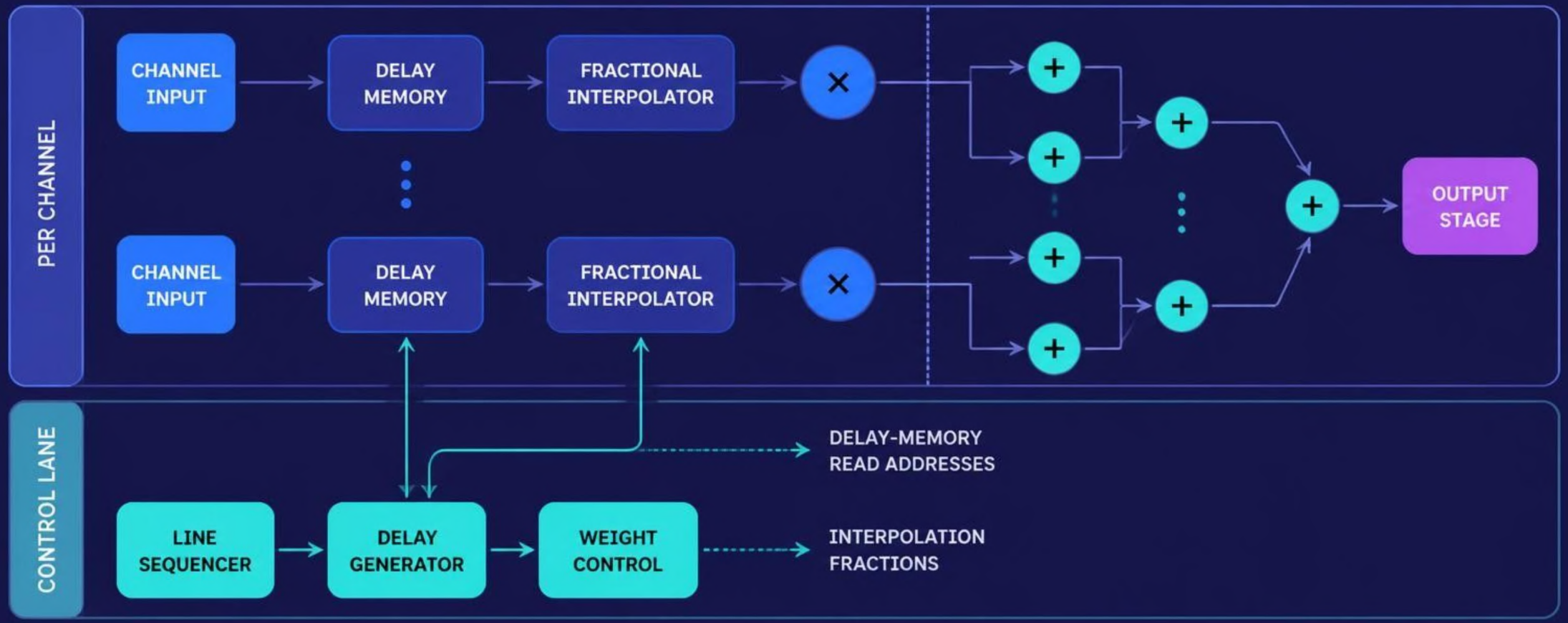
Pipelines deliver deterministic, low-latency processing.

Local resources

DSP slices and block RAM support multiply, delay, and sum.

Large devices offer thousands of compute and memory resources; capacity depends on the selected FPGA.

Beamformer architecture



Getting ADC data into the FPGA

Recover

Use LVDS deserializers or JESD204 transceivers.

Align

Establish clock, word, lane, and channel alignment.

Validate

Check test patterns before starting the channel pipeline.

Correct arithmetic cannot repair misframed input samples.

Generating delays: LUT versus on-the-fly

Pre-computed LUT

Simple, predictable readout.

Memory grows with channels, depths, and scan lines.

On-the-fly

Compute geometry or update incrementally.

Less table storage, more arithmetic and control complexity.

Trade memory capacity against logic, throughput, and verification effort.

Fractional delays with interpolation

$$d = k + \mu, \qquad 0 \le \mu < 1$$

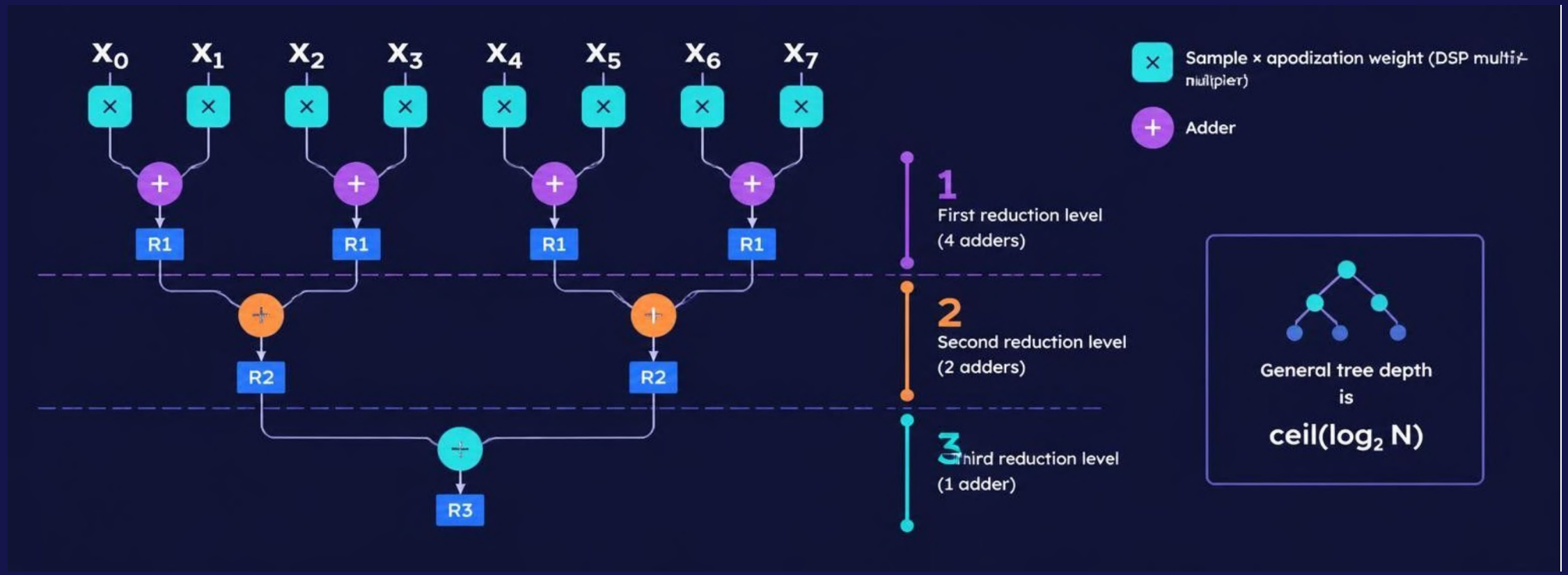
$$\hat{x}[n - d] = (1 - \mu)x[n - k] + \mu x[n - k - 1]$$

The integer part selects buffer samples. The fraction blends adjacent samples, or selects a longer fractional-delay FIR.

Improve focus without increasing ADC sample rate.

Linear interpolation is inexpensive; FIR interpolation offers better bandwidth fidelity at higher cost.

Apodization and summation



DSP slices multiply samples by weights. A balanced, registered tree reduces all channels to one sum.

Pipelining and timing closure

Register

Register every stage of the adder tree to bound logic depth.

Clock

Keep the clock at the ADC sample rate or a multiple.

Latency

Latency must be known and constant for a fixed configuration.

After the sum: demodulation and decimation

Mix

Translate RF into baseband I/Q.

Low-pass

Retain the useful echo bandwidth and reject images.

Decimate

Reduce sample rate only after anti-alias filtering.

Send less data to the host while preserving the information needed for imaging.

A prototyping workflow that works

Golden model

Build MATLAB or Python reference processing first.

RTL comparison

Simulate identical inputs and compare stage outputs.

Hardware

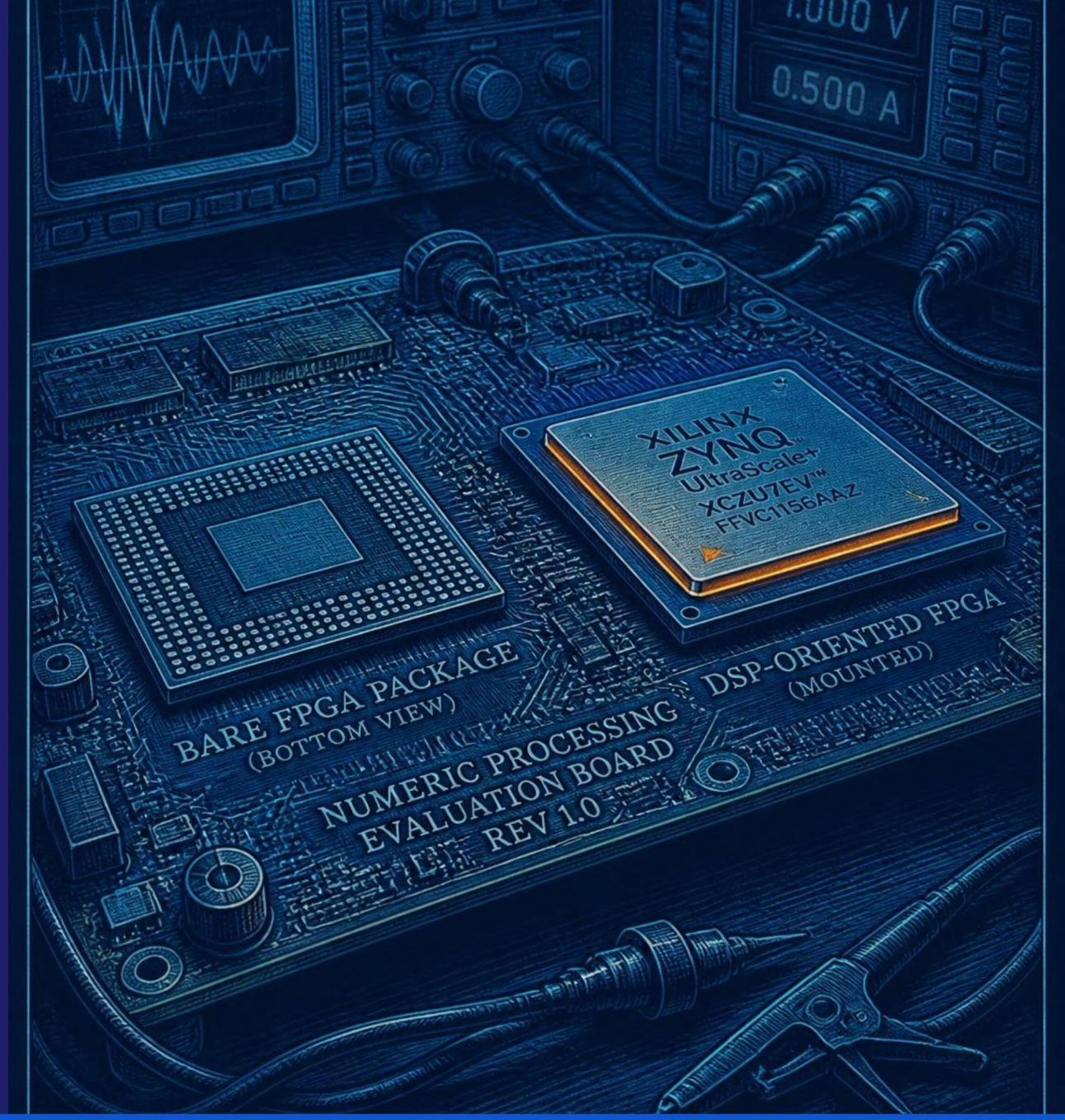
Bring up an ADC evaluation platform, then instrument real streams.

Keep captured inputs reproducible so every mismatch can be replayed.

PART 6

Fixed point vs floating point

Precision is an architectural resource.



Fixed-point numbers

An integer with an implied binary point. Qm.n assigns integer and fractional bits; state whether m includes the sign.



$$\text{value} = \text{stored signed integer} \times 2^{-n}$$

Fixed range. Constant step size. Small and fast arithmetic.

Floating-point numbers

IEEE 754 separates sign, exponent, and fraction. The exponent moves the effective binary point.



IEEE 754 binary32, commonly called single precision. Normal values have an implicit leading significand bit.

Huge range and relative precision, with more arithmetic logic.

Fixed versus floating at a glance

	Fixed point	Floating point
Range	Limited	Very wide
Precision	Constant step	Relative
FPGA cost	Low	Higher
Speed	High	Lower unless hard FP
Design effort	Bit-width analysis	Simpler to write

Actual throughput and resource use depend on device, IP, and pipeline configuration.

Why it matters for beamforming RTL

Dynamic range

Weak and strong echoes must coexist.

Accumulation

Many coherent channels increase peak magnitude.

Failure modes

Too little range clips. Too little precision adds noise.

Budget bits before writing RTL.

Bit growth through the beamformer

01

12 bits

ADC sample

Multiply by a 16-bit apodization coefficient.

02

28 bits

Full product

Retain the full multiplication result.

03

35 bits

128-channel sum

Seven guard bits cover worst-case coherent growth.

Track binary-point position separately from width. Narrow only where the error budget permits.

Rounding, saturation, and quantization noise

Round

Truncation can introduce bias. Defined rounding reduces systematic error.

Saturate

Clamp overflow instead of wrapping into the opposite sign.

Measure

An ideal quantizer gains about 6 dB SNR per added bit.

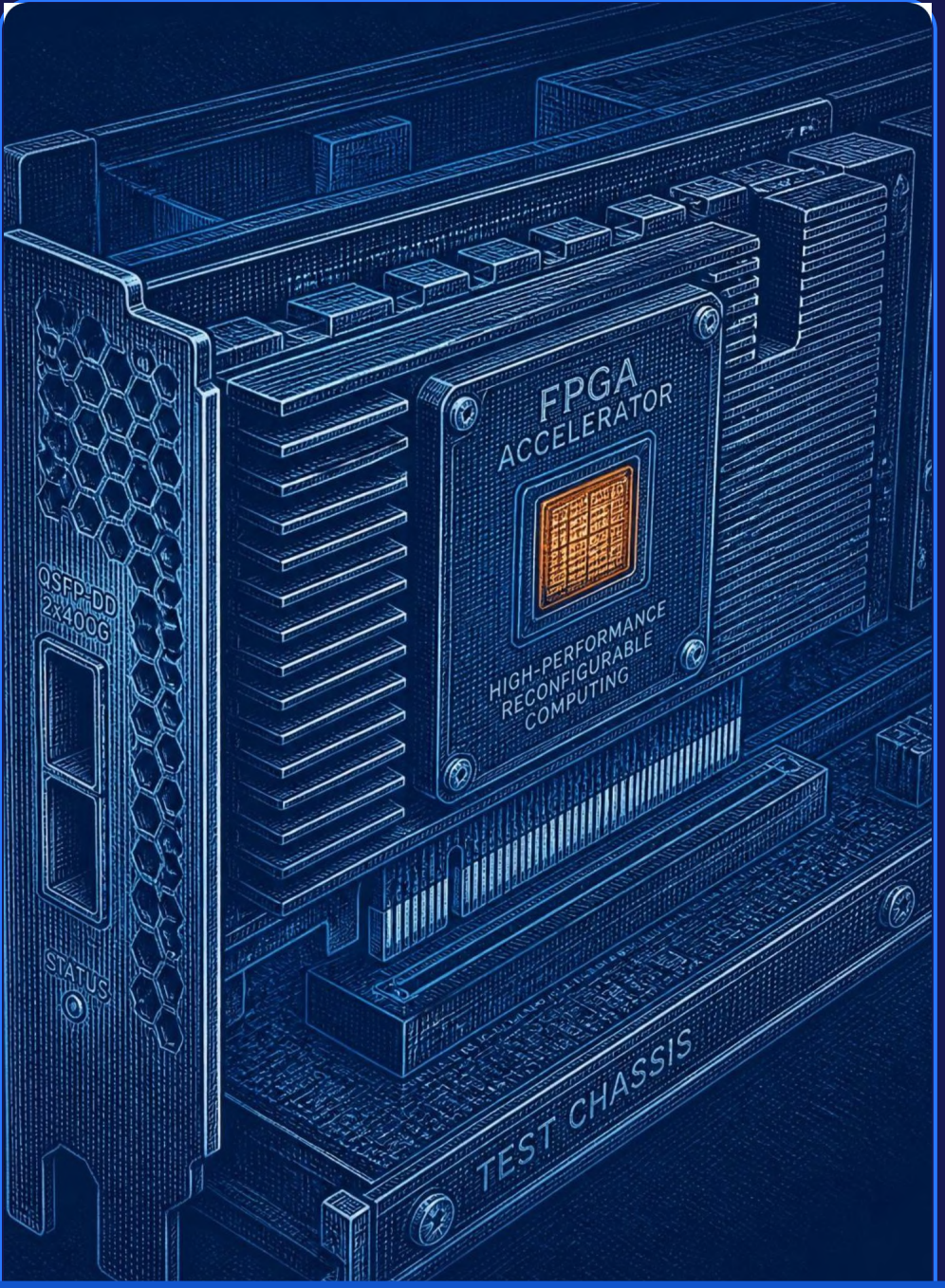
The 6 dB rule assumes ideal quantization and a suitable full-scale signal; system noise may dominate.

Fixed-point multiply-accumulate in RTL

Use signed operands and explicit widths. Preserve the product, sign-extend into the accumulator, then round and saturate at the output boundary.

Signal	Signed width
sample	[11:0]
weight	[15:0]
product	[27:0]
accumulator	[34:0]

assign product = sample * weight. On each posedge clk, clear sets accumulator to '0; valid accumulates accumulator + {{7{product[27]}}, product}.



Floating point on FPGA

Vendor IP provides pipelined floating-point operators, often with many cycles of latency.

Some devices include hard floating-point DSP blocks.

Useful for prototypes, scan conversion, and selected Doppler calculations.

Check initiation interval as well as latency.

Choosing and validating the number format

Allocate

Prefer fixed point in the high-rate channel path.

Relax

Use float where range matters and rate or resource budget permits.

Prove

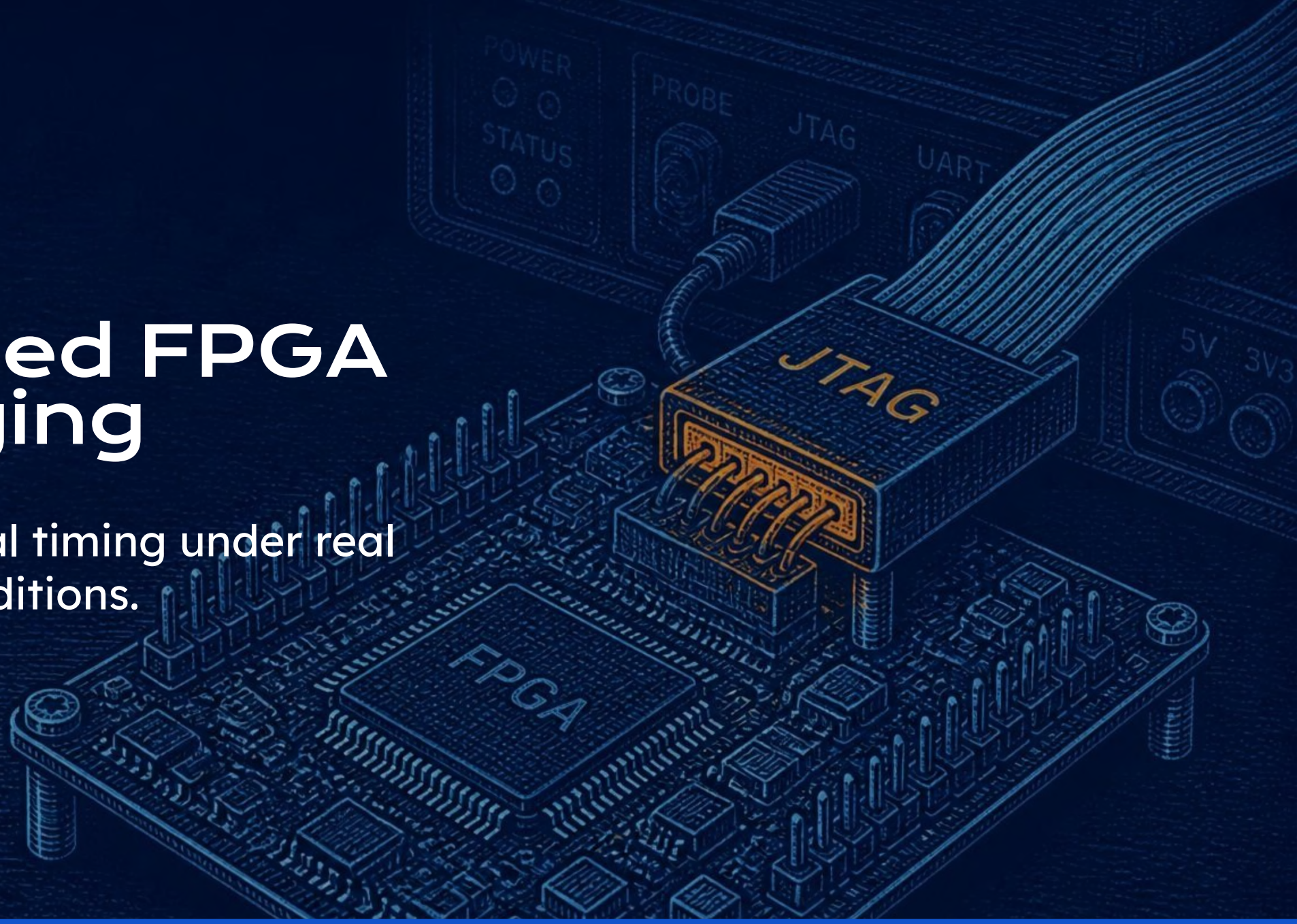
Compare RTL against the floating-point golden model.

Test weak echoes, strong reflectors, coherent sums, and overflow boundaries.

PART 7

Advanced FPGA debugging

Observe internal timing under real acquisition conditions.



Why ultrasound designs need hardware debug

Real inputs

Simulation cannot reproduce every ADC and probe behavior.

Hidden faults

Clocking, channel alignment, and overflow can fail only in hardware.

No probe pins

Important internal signals never reach the board edge.

Instrument the boundaries where a wrong sample first becomes observable.

ILA: the logic analyzer inside the FPGA

01

Sample

Integrated Logic Analyzer probes observe internal signals.

02

Capture

On-chip block RAM stores a finite window of clocked samples.

03

Inspect

Vivado Hardware Manager displays captured waveforms.

ILA and VIO are AMD/Xilinx tool names. Sample each probe in a valid clock domain.

Triggers and capture depth

Find the event

Trigger on a pattern, edge, or sequence.

Useful targets include line start, overflow, and alignment loss.

Keep the context

Choose capture depth and pre-trigger position.

More samples and wider probes consume more BRAM.

$$T_{\text{window}} = \frac{\text{capture samples}}{f_{\text{ILA}}}$$

Adding an ILA to your design

Instrument

Instantiate IP or mark selected nets for debug.

Connect

Attach probes and a stable sampling clock.

Deploy

Rebuild, program, and open the hardware session.

Confirm timing still closes after debug insertion.

VIO: control and watch signals live

Virtual Input/Output exchanges low-rate controls and status with the PC, without rebuilding the bitstream.

PROBE_OUT → DESIGN

Drive controls

Focus depth, gain, apodization, or debug channel selection.

DESIGN → PROBE_IN

Read status

Lock state, counters, configuration acknowledgment, and error flags.

Synchronize controls across clock domains and apply configuration changes at safe boundaries.

ILA and VIO together on the beamformer



Debug best practices

Plan early

Choose few, meaningful probes. Include valid and line or sample identity.

Budget impact

Watch routing, timing, sampling clocks, and BRAM consumption.

Clean release

Remove debug cores from production and revalidate the release build.

Capture evidence that can distinguish competing explanations.

Key takeaways

Beamforming turns channel echoes into an image

FPGAs give parallel, deterministic receive beamforming

Number format choices shape quality and cost

ILA and VIO make real-hardware bugs visible

DISCUSSION

Questions and discussion

